$P(z_1 < Z < 0) = 0.25$ where $z_1 = \frac{x_1 - \mu}{\sigma} = \frac{x_1 - 60}{10}$

From Normal Probability table

$P(-0.675 < Z < 0) = 0.25015$

$\therefore \quad z_1 = -0.675$

$\therefore \quad \frac{x_1 - \mu}{\sigma} = -0.675$

$\frac{x_1 - 60}{10} = -0.675$

$X_1 - 60 = -6.75$

$X_1 = 53.25$ that is $Q_1 = 53.25$

Let x_2 be the value, so that

$\therefore \quad P(\mu < X < x_2) = 0.25$

$P(0 < Z < z_2) = 0.25$, where $z_2 = \frac{x_2 - \mu}{\sigma} = \frac{x_2 - 60}{10}$

From the table

$P(0 < Z < 0.675) = 0.25015$

$\therefore \quad Z_2 = 0.675$

$\frac{x_2 - 60}{10} = 0.675$

$X_2 = 66.75$ that is $Q_2 = 66.75$

Example 10.30

The height of the rose plants in a garden is Normally distributed with a mean 100cms. Given that 10% of the plants have height greater than104cm. Find (i) The S.D of the distribution (ii) The number of plants have height greater than 105cms if there were 200 plants in the garden.

Solution:

(i) The S.D of the distribution

Let X be the height of the rose plants that is Normally distributed.

Given: $P(100 < x < 104) = 40\ \%$

$P(0 < Z < 4/\sigma) = 0.40$ ... (1)